A LINEAR EQUIVALENCE OF DERIVED CATEGORIES WITH CONCENTRATED SOURCE AND K-FLAT TARGET COMES FROM A BIMODULE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-dga/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the choice N = F(A) as a property-(P) module, the endomorphism dga E with H^0(E) = A, the degree-wise truncated subalgebra E' with its quasi-isomorphism zigzag, the equivalence chain with the quasi-inverse identification M maps to M tensor_A^L A, and the closing composition through lemma-compose-tensor-functors are all retained verbatim.

Editorial note. The label proposition-rickard names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own proposition environment in the chapter source.

Allowed standard prerequisites (editorial).

- The endomorphism differential graded algebra of the image of the unit. Same chapter, Remark remark-lift-equivalence-to-dga, Lemma lemma-tilting-equivalence and Lemma lemma-hom-derived. Sets N = F(A) as a property-(P) module, identifies (E, d) = Hom(N, N) with H^0(E) = A and vanishing higher cohomology, and gives through the tilting lemma that N as an (E, B)-bimodule induces an equivalence D(E, d) to D(B, d).
- Derived categories of quasi-isomorphic algebras and restriction of scalars. Same chapter, Lemma lemma-qis-equivalence and Example example-map-hom-tensor. Turns the quasi-isomorphism zigzag of algebras into equivalences of derived categories and identifies the quasi-inverse of restriction with derived tensor.

Source reference. Original reference 0BYX: sources/stacks-dga/pinned-original-source.tex, source lines 5657–5657. Prerequisite pointer invoked at proof line(s) 6950; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09S4: sources/stacks-dga/pinned-original-source.tex, source lines 6067–6067. Prerequisite pointer invoked at proof line(s) 6953; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09KY: sources/stacks-dga/pinned-original-source.tex, source lines 2850–2850. Prerequisite pointer invoked at proof line(s) 6923; it names a standard fact of the same chapter that the counted argument

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uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09S6: sources/stacks-dga/pinned-original-source.tex, source lines 6725–6725. Prerequisite pointer invoked at proof line(s) 6946; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09S7: sources/stacks-dga/pinned-original-source.tex, source lines 6753–6753. Prerequisite pointer invoked at proof line(s) 6912; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09S9: sources/stacks-dga/pinned-original-source.tex, source lines 6865–6865. Prerequisite pointer invoked at proof line(s) 6916; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *Let R be a ring. Let (A, d) and (B, d) be differential graded R-algebras. Let $F : D(A, d) \to D(B, d)$ be an R-linear equivalence of triangulated categories. Assume that*

(1) *$A = H^0(A)$, and*
(2) *B is K-flat as a complex of R-modules.*

Then there exists an (A, B)-bimodule N as in Lemma 09S7.

Proof. As in Remark 09S9 above, we set $N = F(A)$ in $D(B, \mathrm{d})$. We may assume that N is a differential graded B-module with property (P). Set

$$(E, \mathrm{d}) = \mathrm{Hom}_{\mathrm{Mod}^{dg}_{(B,\mathrm{d})}}(N, N)$$

Then $H^0(E) = A$ and $H^k(E) = 0$ for $k \neq 0$ by Lemma 09KY. Moreover, by the discussion in Remark 09S9 and by Lemma 09S7 we see that N as a (E, B)-bimodule induces an equivalence $- \otimes^{\mathbf{L}}_E N : D(E, \mathrm{d}) \to D(B, \mathrm{d})$. Let $E' \subset E$ be the differential graded R-subalgebra with

$$(E')^i = \begin{cases} E^i & \text{if } i < 0 \\ \mathrm{Ker}(E^0 \to E^1) & \text{if } i = 0 \\ 0 & \text{if } i > 0 \end{cases}$$

Then there are quasi-isomorphisms of differential graded algebras $(A, \mathrm{d}) \leftarrow (E', \mathrm{d}) \to (E, \mathrm{d})$. Thus we obtain equivalences

$$D(A, \mathrm{d}) \leftarrow D(E', \mathrm{d}) \to D(E, \mathrm{d}) \to D(B, \mathrm{d})$$

by Lemma 09S6. Note that the quasi-inverse $D(A, \mathrm{d}) \to D(E', \mathrm{d})$ of the left vertical arrow is given by $M \mapsto M \otimes^{\mathbf{L}}_A A$ where A is viewed as a (A, E')-bimodule, see Example 0BYX. On the other hand the functor $D(E', \mathrm{d}) \to D(B, \mathrm{d})$ is given by $M \mapsto M \otimes^{\mathbf{L}}_{E'} N$ where N is as above. We conclude by Lemma 09S4. □